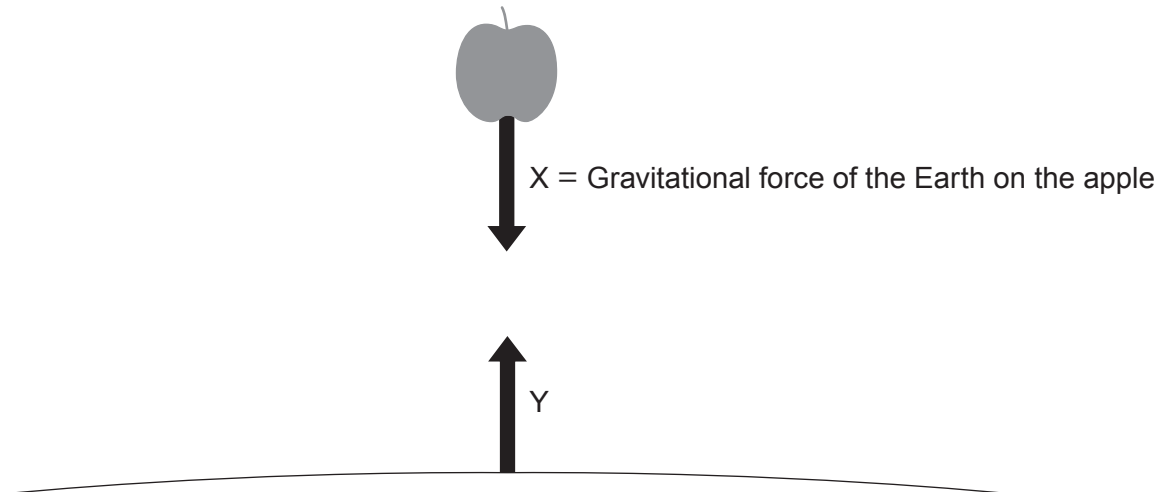


3. Newton's third law states that if a body A exerts a force on body B, body B exerts an equal and opposite force on body A.

(a) The diagram (not to scale) shows an apple of weight 1.5 N falling to the Earth.



- (i) Underline the correct word in the brackets to describe force Y. [2]

Force Y is the (**gravitational** / magnetic / mass) force of the apple on the (air / **Earth** / Moon).

- (ii) State how the two forces, X and Y, compare. [1]

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- (iii) Use an equation from page 2 to calculate the mass of the apple. [2]
($g = 10 \text{ N/kg} = 10 \text{ m/s}^2$)

mass = kg

(b) The apple is dropped from rest.

- (i) State the initial acceleration of the apple. [1]

acceleration = m/s^2



- (ii) At some time later in the fall, the air resistance acting on the apple is 0.25 N.



Determine the size of the resultant force acting on the apple.

[1]

resultant force = N

- (iii) Use the equation:

$$\text{resultant force} = \text{mass} \times \text{acceleration}$$

to calculate the new acceleration of the apple.

[3]

acceleration = m/s^2

- (c) The Apollo 15 astronaut David Scott performed an experiment on the Moon, where there is no air.

He predicted that if a hammer and a feather were dropped together from the same height, they would hit the Moon's surface at the same time.

Explain whether you agree with this prediction.

[2]

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